

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

ME 2127
Solid Mechanics

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Explain the following terms: (08)
(i) Hook's law (ii) Poisson's ratio (iii) Yield strength and (iv) Rupture strength
(b) Show that the tangential stress in a thin-walled cylindrical shell of diameter 'D' and wall thickness 't' subjected to internal pressure 'P' is given by (12)
$$\sigma_t = PD/2t$$

(c) An aluminium rod is rigidly attached between a steel rod and a bronze rod as shown in Figure: 1(c). Axial loads are applied at the positions indicated. Find the maximum value P that will not exceed a stress in steel of 140 MPa, in aluminium of 90 MPa, or in bronze of 100 MPa. (15)
- Q 2. (a) A uniform concrete slab of total weight W is to be attached, as shown in Figure: 2(a), to two rods whose lower ends are on the same level. Determine the ratio of the areas of the rods so that the slab will remain level. (15)
(b) Write the shear and moment equations and draw shear and moment diagrams for the beam loaded as shown in Figure: 2(b). Specify values at all changes of loading positions and at all points of zero shear. (20)
- Q 3. (a) A compound shaft consisting of a steel segment and an aluminium segment is acted by upon two torques as shown in Figure: 3(a). Determine the maximum permissible value of T subject to the following conditions: $\tau_{st} \leq 83$ MPa, $\tau_{al} \leq 55$ MPa, and the angle of rotation of the free end is limited to 6°. For steel, $G = 83$ GPa and for aluminium, $G = 28$ GPa. (17)
(b) A rigid bar, hinged at one end, is supported by two identical springs as shown in Figure: 3(b). Each spring consists of 20 turns of 10 mm wire having a mean diameter of 150 mm. Compute the maximum shearing stress in the springs, neglecting the mass of the rigid bar. (18)
- Q 4. (a) Derive the flexural formula, $\sigma = \frac{My}{I}$ mentioning appropriate assumptions, where symbols have their usual meanings. (17)
(b) Compute the values of $EI\delta$ at midspan for the beam loaded as shown in Figure: 4(b). If $E = 10$ GPa, what value of I is required to limit the midspan deflection 1/360 of the span? (18)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) A wooden beam 100 mm by 200 mm, supported as shown in Figure: 5(a), carries a load P. (17)
What is the largest safe value of P if the maximum is not to exceed 10 MPa?
- (b) If an element is subjected to the state of stress shown in Figure: 5(b), find the principal stresses (18)
and maximum in plane shearing stress.
- Q 6. (a) Determine the force in each member of the truss [shown in Figure: 6(a)] and state if the (18)
members are in tension or compression. Set, $P_1=100$ lb, $P_2=200$ lb, $P_3=300$ lb.
- (b) Determine the components of the forces acting on each member of the frame shown in (17)
Figure: 6(b).
- Q 7. (a) Determine the centroid $(\bar{x}, \bar{y})$ of the shaded area as shown in Figure: 7(a). (18)
- (b) Determine the distance $\bar{z}$ to the centroid of the shape [shown in Figure: 7(b)] that consists of a (17)
cone with a hole of height $h=50$ mm bored into its base.
- Q 8. (a) Determine the moment of inertia of the shaded area [shown in Figure: 8(a)] with respect to x (17)
and y axis.
- (b) Determine the moment of inertia of the area [shown in Figure: 8(b)] with respect to x and y (18)
axis.

Section A

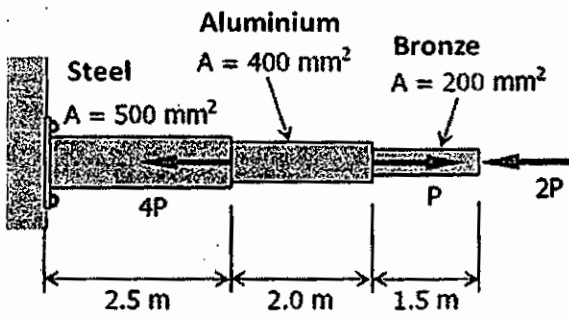


Figure: 1(c)

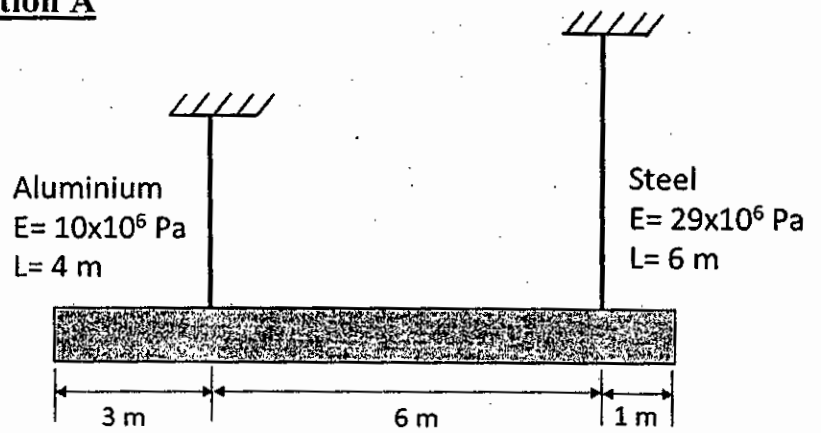


Figure: 2(a)

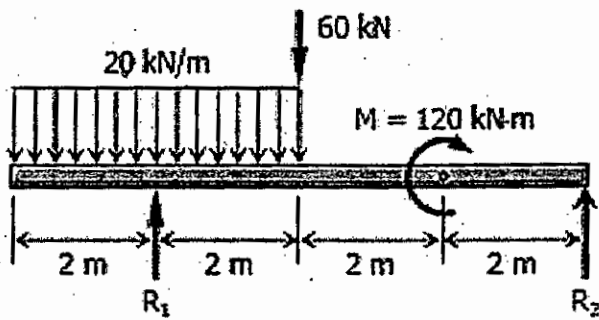


Figure: 2(b)

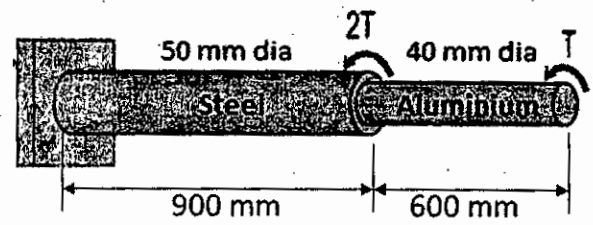


Figure: 3(a)

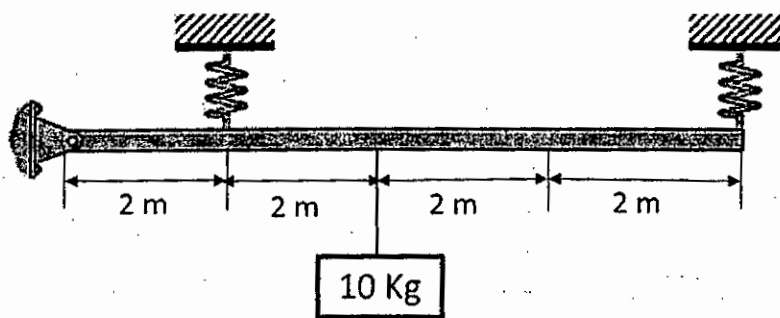


Figure: 3(b)

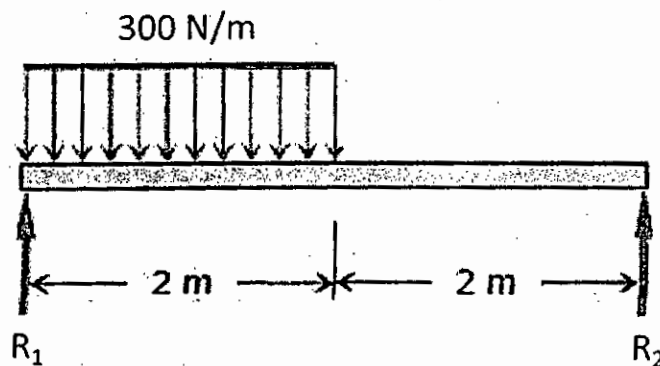


Figure: 4(b)

Section B

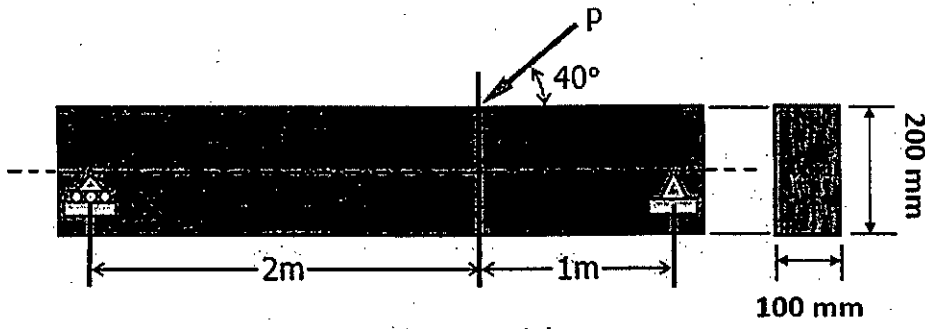


Figure: 5(a)

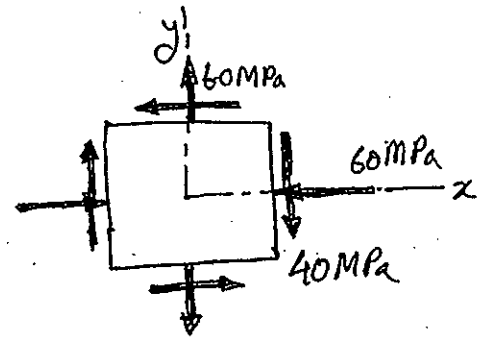


Figure: 5(b)

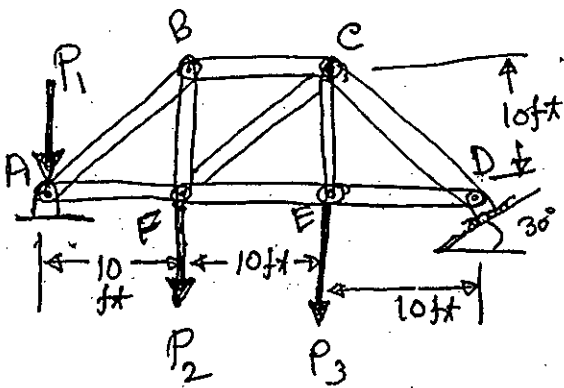


Figure: 6(a)

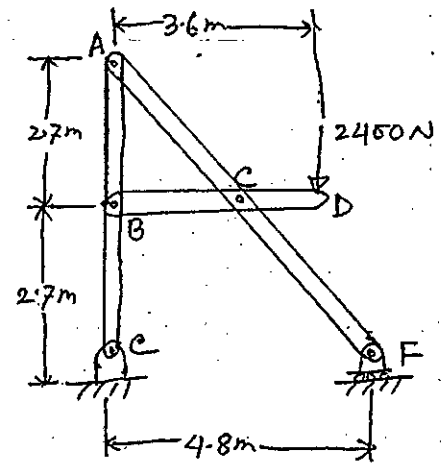


Figure: 6(b)

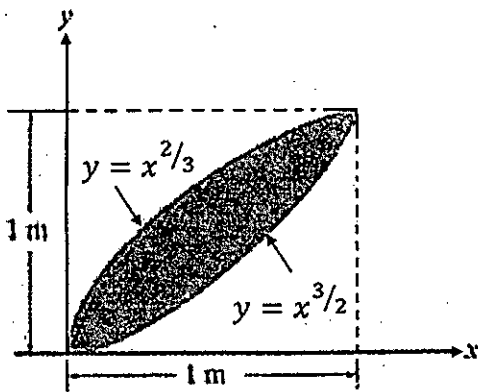


Figure: 7(a)

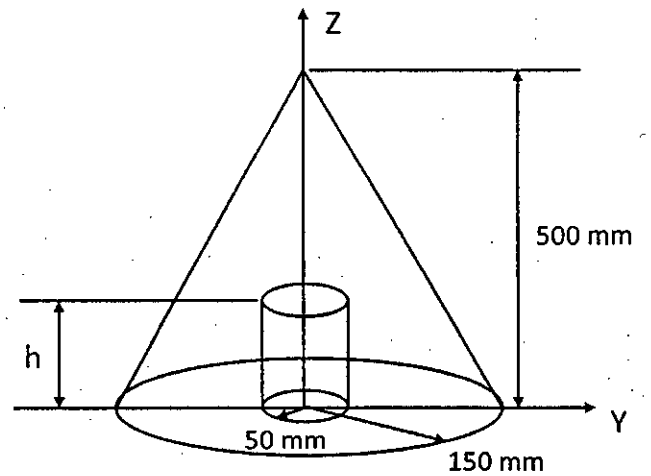


Figure: 7(b)

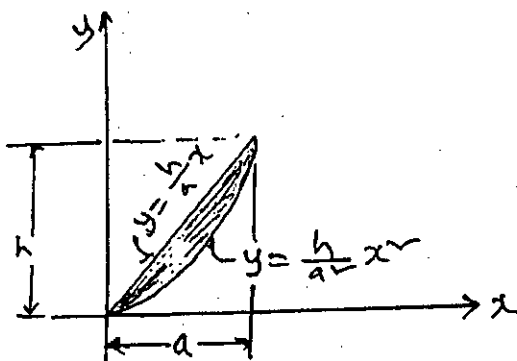


Figure: 8(a)

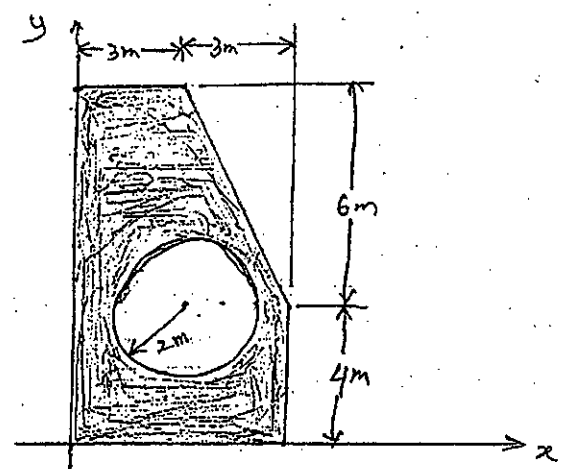


Figure: 8(b)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

MSE 2103
Thermodynamics of Materials

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Write down the three laws of thermodynamics. (06)
(b) Explain the concepts of reversible process and irreversible process with a real-life example. (15)
(c) Prove that, heat absorbed in any reversible isothermal process is the maximum. (08)
(d) Describe an experimental validation of the third law of thermodynamics. (06)
- Q 2. (a) Derive the combined statement of the first law and the second law of thermodynamics for a simple system. (07)
(b) Derive an expression for the increase in temperature for a process in which the volume of the system is changed at constant entropy. (12)
(c) One mole of nickel initially at 300 K and 1 atm pressure is taken through two separate process: (16)
(i) An isobaric change in temperature to 1000 K.
(ii) An isothermal compression to 1000 atm.
Compute the change in enthalpy of nickel for these two processes. [N.B. derive exact results]
Use the following data in your calculations:
 $V(300\text{ K}, 1\text{ atm}) = 6.57\text{ cc/mol}$
 $\alpha = 40 \times 10^{-6}\text{ K}^{-1}$
 $\beta = 26 \times 10^{-7}\text{ K}^{-1}$
 $C_p = 11.17 + 37.78 \times 10^{-3}T + 3.18 \times 10^5 T^{-2}\text{ J/mol} - K$
- Q 3. (a) Derive the conditions for equilibrium for a two-phase unary system using a constrained maximum of entropy. (20)
(b) Differentiate between steady state and equilibrium state with an example. (08)
(c) "The entropy of the universe can only increase"- Explain (07)
- Q 4. (a) Deduce the general equations from which the temperature (T_i) and the pressure (P_i) at any triple point in a unary phase diagram can be computed. (15)
(b) Show that, in an open unary system, chemical potential is identical to Molar Gibbs free energy. (08)
(c) Point out the characteristics of unary phase diagrams in (P, T) space. (07)
(d) "Slopes of the vapor pressure curves of the elements increase with increasing boiling points of the elements"- Why? (05)

Section B

(Answer ANY THREE questions from this section in **answer script B**)

- Q 5. (a) What is phase? Determine the equilibrium condition of an isolated system having P phases and C components. (20)

$$[\text{Given, } dU' = TdS' - PdV' + \sum_{k=1}^C \mu_k dn_k]$$

- (b) What is degree of freedom? Derive the equation for Gibbs phase rule where the system consists of P phases and C components. (15)

- Q 6. (a) Deduce an expression for the Clausius-Clapeyron equation mentioning all the assumptions to be made in establishing the relation. (20)

- (b) The normal boiling point of zinc is 1180 K. Calculate its vapor pressure at 1073 K, assuming that zinc follows Trouton's rule. (15)

- Q 7. (a) Explain the concept of partial molar properties. How would you evaluate partial molar properties of a component in a binary solution knowing the partial molar properties of others? (20)

$$[\text{Provided, } d\Delta B_{mix} = \Delta \bar{B}_1 dX_1 + \Delta \bar{B}_2 dX_2]$$

- (b) Compute the partial molar enthalpies of the two components in a binary solution with an enthalpy of mixing given by $\Delta H_{mix} = aX_1X_2$ (15)

- Q 8. (a) Out of C and CO, which one is the better reducing agent for the reduction of FeO? Use Ellingham diagram to give your answer. [See Figure: 8(a)] (10)

- (b) With the help of Ellingham diagram, explain why lead (Pb) is not used as a reducing agent in the manufacturing of chromium from chromium oxide. [See Figure: 8(a)] (10)

- (c) Draw the free energy-composition diagrams of the following phase diagram at the indicated temperatures. [See Figure: 8(c)] (15)

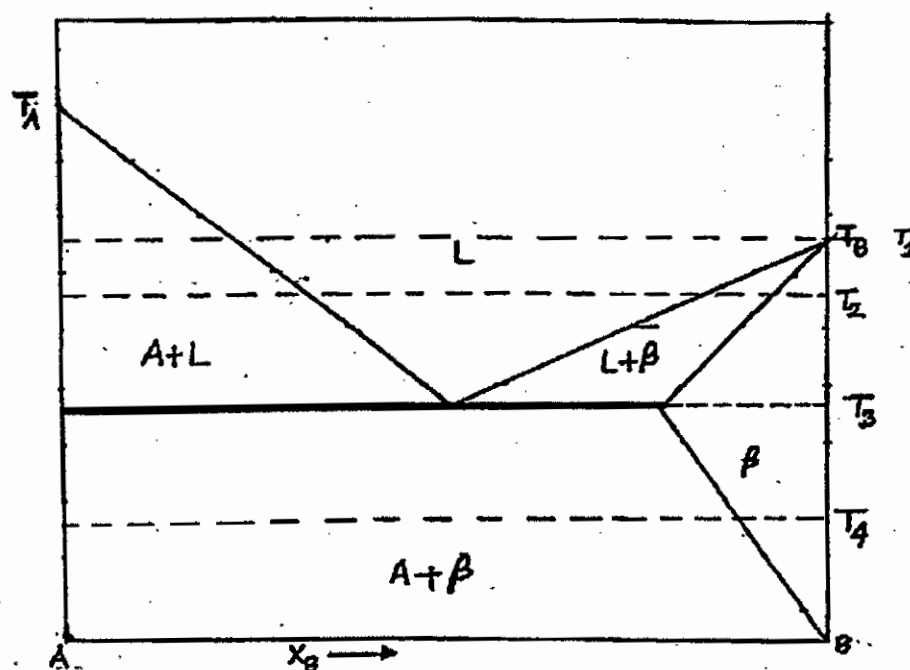


Figure: 8(c)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

MSE 2101
Crystallography

Time: 3 hours

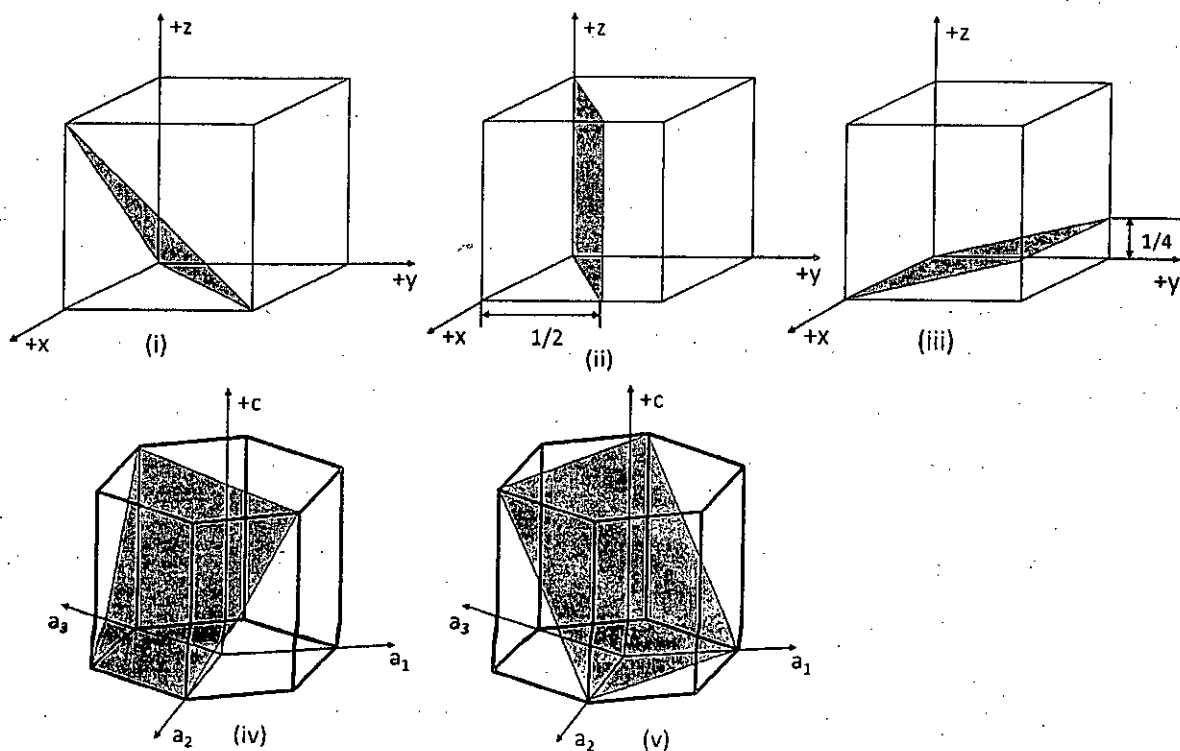
Full Marks: 210

- N.B. (i) Answer ANY THREE questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer ANY THREE questions from this section in answer script A)

- Q 1. (a) How many types of interstitial sites are available in different atomic-stacking? Calculate the number of tetrahedral and octahedral voids surrounding a sphere of close packed stacking. (15)
- (b) Define space and crystal lattice with proper example. (10)
- (c) What is unit cell? "The choice of unit cell is not unique" - Explain with appropriate example. (10)
- Q 2. (a) What is symmetry? What are the differences between trigonal and hexagonal crystal structures, though they possess similar lattice parameters of $a=b \neq c$, $\alpha=\beta=90^\circ$, $\gamma=120^\circ$? (15)
- (b) Prove that, the possibility of five, seven, eight (and so on) fold rotational axes symmetry are not possible in an ideal crystal structure. (10)
- (c) On the basis of symmetry, a monoclinic crystal ($a \neq b \neq c$, $\alpha=\gamma=90^\circ$) may be primitive, P or C-centered. Are other distinct types of centering possible? Justify your answer. (10)
- Q 3. (a) What are the differences between FCC and HCP crystal lattice though both of them are constructed from a close packed stacking? (10)
- (b) Calculate the ideal c/a ratio of HCP crystal and using the relationship, develop an equation for packing density of the HCP crystal. (15)
- (c) Determine the Miller indices of the following crystallographic planes: (10)



- Q 4. (a) Explain the reasons behind four-coordinate system for hexagonal crystal system instead of conventional three-coordinate system. (11)
- (b) Explain, why a larger interstitial atom preferentially accommodate in the tetrahedral site rather than octahedral site of a BCC crystal. (12)
- (c) Describe the zinc blende structure with neat sketch and justify such coordination configuration. (12)
- (Given, ionic radius of Zn and S are 74 pm and 184 pm, respectively).

Section B

(Answer ANY THREE questions from this section in **answer script B**)

- Q 5. (a) Does diffraction occur in amorphous materials? Explain your answer. (05)
- (b) Derive Bragg's law for diffraction, and show that, a reflection of any order can be treated as a first order reflection. Clarify why Bragg's law is a necessary but not sufficient condition for diffraction by real crystals. (15)
- (c) What are the three scattering modes experienced by atoms in different materials? (05)
- (d) Using a diffractometer and a radiation of wavelength $1.54 \text{ \AA}$, a reflection from an FCC crystal is observed when 2θ is 121° . Calculate the indices (hkl) of this reflection. What is the spacing between planes of the crystal from which this reflection originates? (10)
- Q 6. (a) With appropriate figures, describe the steps involved in obtaining a two-dimensional reciprocal lattice. (15)
- (b) If a plane (hkl) lies in a zone [uvw], show that, (i) $hu + kv + lw = 0$, and (ii) $hu + kv + lw = n$, for n^{th} parallel plane from the plane (hkl). (10)
- (c) Draw the plan view (with Miller indices) of a BCC crystal perpendicular to the z-axis. From this plan view, draw the pattern of reciprocal lattice points perpendicular to the z-axis. (10)
- Q 7. (a) What is "foreshortening"? Clarify with an example. Describe how foreshortening is eliminated in stereographic projection. (13)
- (b) Explain how you would determine the angle between crystallographic planes using Wulff net. (10)
- (c) Provide proper examples of the operations of the following point group symmetry elements using stereographic projection: (i) mirror planes, (ii) rotational symmetry, and (iii) inversion axis of symmetry. (12)
- Q 8. (a) Distinguish between diffraction of X-rays and reflection of visible light. (09)
- (b) Describe the working principle of an X-ray diffractometer using a schematic diagram. Explain how variation of diffraction angle is obtained in powder diffraction. (14)
- (c) Mention the reasons of peak-shifting in X-ray diffraction. (06)
- (d) X-rays with a wavelength of $0.58 \text{ \AA}$ are used for calculating d_{200} in nickel. The reflection angle is 9.5° . What is the edge length of the unit cell? (06)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

Math 2127
Vector Analysis and Linear Algebra

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer ANY THREE questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer ANY THREE questions from this section in answer script A)

- Q 1. (a) A particle moves along the curve $x=t^3+1, y=t^2, z=2t+5$, where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $2\mathbf{i}+3\mathbf{j}+6\mathbf{k}$. (10)
- (b) Motion of a satellite is described by $\frac{d^2\mathbf{r}}{dt^2}+2\frac{d\mathbf{r}}{dt}+5\mathbf{r}=\mathbf{0}$. Determine the position of it for any time t . (12)
- (c) Find the angle between the surfaces $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at the point $(2, -1, 2)$. (13)
- Q 2. (a) The temperature at any point in space is given by $T=xy+yz+zx$. Determine the derivative of T in the direction of the vector $3\mathbf{i}-4\mathbf{k}$ at the point $(1, 1, 1)$. (10)
- (b) If $\nabla \cdot \mathbf{E}=0, \nabla \cdot \mathbf{H}=0, \nabla \times \mathbf{E}=-\frac{\partial \mathbf{H}}{\partial t}, \nabla \times \mathbf{H}=\frac{\partial \mathbf{E}}{\partial t}$, then show that the electric field $\mathbf{E}$ and the magnetic field $\mathbf{H}$ satisfy the wave equation $\nabla^2 u = \frac{\partial^2 u}{\partial t^2}$. (10)
- (c) What is meant by a conservative force field? Show that $\mathbf{F}=(2xy+z^3)\mathbf{i}+x^2\mathbf{j}+3xz^2\mathbf{k}$ is a conservative force field. Also find the scalar potential and hence find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$. (15)
- Q 3. (a) What is the relation between line integral and surface integral? Evaluate $\iint_S \mathbf{A} \cdot \mathbf{n} dS$, where $\mathbf{A}=(x+y^2)\mathbf{i}-2xz\mathbf{j}+2yz\mathbf{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant. (15)
- (b) State the divergence theorem of Gauss. Find $\iint_S \mathbf{F} \cdot \mathbf{n} dS$, where $\mathbf{F}=4xz\mathbf{i}-y^2\mathbf{j}+yz\mathbf{k}$ and S is the surface of the cube bounded by $x=0, x=1, y=0, y=1, z=0, z=1$. (10)
- (c) Verify Green's theorem for $\int_C (x^2-2xy)dx+(x^2y+3)dy$ around the boundary C of the region $y^2=8x$ and $x=2$. (10)
- Q 4. (a) Represent the vector $\mathbf{A}=z\mathbf{i}-2xz\mathbf{j}+y\mathbf{k}$ in cylindrical coordinates. Thus, determine A_ρ, A_ϕ and A_z . (13)
- (b) The acceleration $\mathbf{a}$ of a rocket at any time $t \geq 0$ is given by $\mathbf{a}=e^{-t}\mathbf{i}-6(t+1)\mathbf{j}+3 \sin t \mathbf{k}$. If the velocity $\mathbf{v}$ and displacement $\mathbf{r}$ are zero at $t=0$, then find $\mathbf{v}$ and $\mathbf{r}$ at any time. (10)
- (c) Express Green's theorem in the plane in vector notation. (6)
- (d) Prove that, for every field $\mathbf{v}$, $\text{div curl } \mathbf{v}=0$. (6)

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) What is meant by homogenous system of linear equations? Find the values of the coefficients a, b, and c so that the following linear system

$$ax+by-3z=-3$$

$$-2x-by+cz=-1$$

$$ax+3y-cz=-3$$

has the solution $x=1, y=-1$ and $z=2$.

- (b) Reduce the matrix A to normal form by the elementary transformations, and hence find its rank, where (1)

$$A = \begin{pmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{pmatrix}$$

- (c) Find the inverse of the matrix B by elementary transformations, where (1)

$$B = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

- Q 6. (a) When is a system of linear equations said to be consistent? For which values of 'a', the following system has exactly one solution: (1)

$$x+2y-3z=4$$

$$3x-y+5z=2$$

$$4x+y+(a^2-14)z=(a+2)$$

- (b) What is symmetric matrix? Find all values of a, b, and c for which A is symmetric. (

$$A = \begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$$

- (c) What is null space? Find a basis for the null space of A where (

$$A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$$

- Q 7. (a) Define subspace. Let S and T be the subspaces defined below. Find the basis and dimension of $S \cap T$, where (

$$S = \{(x, y, z, t) : y+z+t=0\}$$

$$T = \{(x, y, z, t) : x+y=0, z=2t\}$$

- (b) If $M = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$, then find all eigenvalues of M and the corresponding eigenvectors. Also evaluate an invertible matrix P such that $P^{-1}MP$ is diagonal.

- (c) Find the coordinate vector of v relative to the basis, $S = \{v_1, v_2, v_3\}$, where

$$v = (5, -12, 3), v_1 = (1, 2, 3), v_2 = (-4, 5, 6), v_3 = (7, -8, 9)$$

- Q 8. (a) Obtain the linear transformation which reduces the quadratic form $12x_1^2 + 4x_2^2 + 5x_3^2 - 6x_1x_2 - 4x_2x_3 + 6x_1x_3$ to the sum of squares and hence reduce it to the canonical form.

- (b) What is meant by the vector space? Determine whether or not the vector $v = (3, 9, -4, -2)$ is a linear combination of the vectors $u_1 = (1, -2, 0, 3)$, $u_2 = (2, 3, 0, -1)$ and $u_3 = (2, -1, 2, 1)$.

- (c) Show that $T = \{(a, b, c) \in \mathbb{R}^3 : 2a - 3b + 5c = 0\}$ is a subspace of $\mathbb{R}^3$.

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B.Sc. Engineering 2nd Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

HUM 2127
Accounting and Economics

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer ANY THREE questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer ANY THREE questions from this section in **answer script A**)

- Q 1. (a) What is conceptual framework of Accounting? Discuss the basic Principles of Accounting. (15)
(b) Why is accounting called an information system? (5)
(c) Define depreciation. Describe the methods of computing depreciation. (15)
- Q 2. (a) What is Accounting cycle? What are the steps of an accounting cycle? (15)
(b). The following accounts are taken from the ledger of Nasir Company at December 31, 2017. (20)

Serial No.	Accounts	Tk.
1	Cash	6,000
2	Notes payable	20,000
3	Nasir, capital	25,000
4	Furniture	80,000
5	Nasir, withdrawal	8,000
6	Depreciation expense	38,000
7	Sales revenue	88,000
8	Equipment	5,000
9	Rent expense	2,000
10	Salaries payable	3,000
11	Accounts payable	11,000
12	Accounts receivable	8,000
13	Unearned revenue	3,000
14	Prepaid insurance	3,000

Requirement: Prepare a trial balance.

- Q 3. (a) Briefly discuss the elements of financial statements. (7)
(b) The information of Sony Trading as of December 31, 2016 are given below: (28)

Account Titles	Tk.
Accounts Receivable	48,400
Interest Revenue	5,000
Prepaid Rent	7,920
Prepaid Insurance	3,960
Equipment	1,32,000

Accumulated Depreciation-Equipment	44,000
Rental Service Revenue	3,80,000
Salaries Expense	2,75,000
Travel Expense	38,940
Interest Expense	13,200
Cash	70,400
Accounts Payable	34,100
Sony, Capital (01.01.16)	1,53,120
Sony, Drawing	26,400

Additional data:

1. The prepaid rent was for the period January 1, 2016 to December 31, 2016.
2. The depreciation on the delivery equipment is tk. 8,800 per year.
3. The prepaid insurance was for the period April 1, 2016 to March 31, 2017.
4. Unpaid salaries tk. 3,300 at December 31.

Requirements: Prepare (i) Income Statement (ii) Statement of owner's Equity (iii) Balance Sheet

- Q 4. (a) Define tax and tax holiday. (8)
- (b) Describe the principles or canons of taxation. (15)
- (c) Define business environment. What are the external factors of business environment? (12)

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Define economics. Distinguish between Micro and Macro economics. (15)
- (b) What are the basic problems of an economic system? (10)
- (c) Discuss the different economic systems. (10)
- Q 6. (a) What are the demand schedule and demand curve and how are they related? (10)
- (b) Distinguish between Perfect competition and Monopoly. (10)
- (c) Explain the equilibrium of a market. (15)
- Q 7. (a) What is indifference curve? Explain the characteristics of indifference curve. (15)
- (b) What is meant by a "competitive Firm"? Under what conditions will a Firm exit a market in long run and shutdown a market in short run? Explain. (20)
- Q 8. (a) Explain the idea of real GDP. List the Four components GDP. Give an example of each. (15)
- (b) Explain the idea of Inflation, Demand pull inflation, and Cost push inflation. (20)